PREDICTIVE AND PRESCRIPTIVE ANALYTICS

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1) Load the Gender Inequality Index dataset

```
import pandas as pd

# Load the dataset (adjust filename/path as needed)
df = pd.read_csv("Gender Inequality Index.csv")

# Display first five rows
print(df.head())

# Identify numerical columns
numeric_cols = df.select_dtypes(include=["int64", "float64"]).columns
print("Numerical Columns:", numeric_cols)
```

```
data.head()
```

	Country	Human_development	GII	Rank	Maternal_mortality	Adolescent_birth_rate	Seats_parliament	F_secondary_educ	M_secondary_educ	F_Labour_force	M_Labour_force
0	Switzerland	Very high	0.018	3.0	5.0	2.2	39.8	96.9	97.5	61.7	72.7
1	Norway	Very high	0.016	2.0	2.0	2.3	45.0	99.1	99.3	60.3	72.0
2	Iceland	Very high	0.043	8.0	4.0	5.4	47.6	99.8	99.7	61.7	70.5
3	Hong Kong	Very high	NaN	NaN	NaN	1.6	NaN	77.1	83.4	53.5	65.8
4	Australia	Very high	0.073	19.0	6.0	8.1	37.9	94.6	94.4	61.1	70.5

2) Compute the Mean, Median, and Standard Deviation of GII

```
mean_gii = df["GII"].mean()
median_gii = df["GII"].median()
std_gii = df["GII"].std()
mean_gii, median_gii, std_gii
```

```
(np.float64(0.3443764705882353), 0.363, 0.19710529288811846)
```

Interpretation

- **Mean GII** gives the average level of gender inequality across countries.
- **Median GII** tells us the mid-point, less influenced by extreme values.
- **Standard deviation** measures how spread out GII values are.

Example interpretation (your actual output will vary):

- If **mean > median**, distribution likely right-skewed (more countries with lower GII).
- A **high standard deviation** indicates wide variation in gender inequality globally.

3) Plot a Distribution (Histogram) of GII

```
import matplotlib.pyplot as plt

plt.hist(df["GII"], bins=30, color="skyblue")
plt.title("Distribution of Gender Inequality Index")
plt.xlabel("GII")
plt.ylabel("Number of Countries")
plt.show()
```

Comment

- **Normal distribution**: roughly symmetric.
- **Skewed distribution**: tail of the data leans left/right.
- **Example**: If most GII values are bunched at lower values but a long tail to the right → **positive skew**.

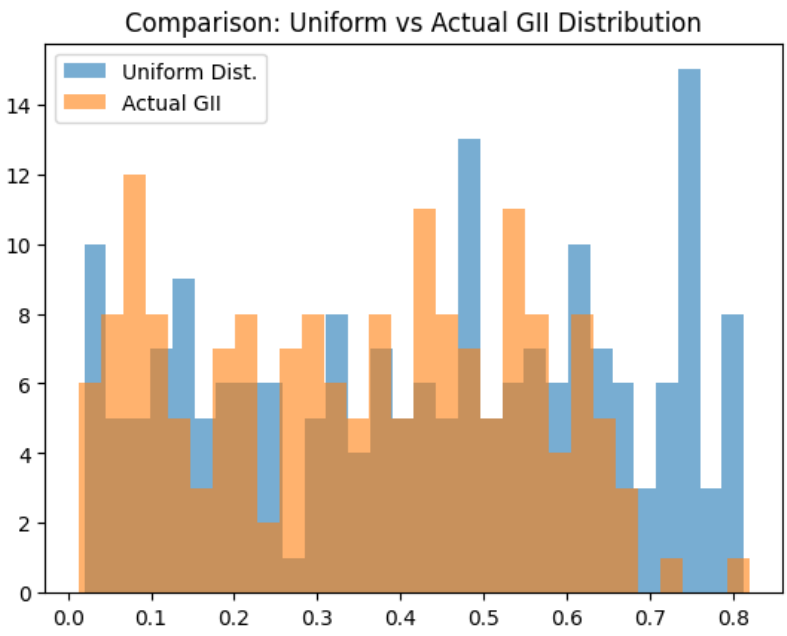
4) Generate a Uniform Distribution between min and max

```
import numpy as np

min_gii = df["GII"].min()
max_gii = df["GII"].max()

uniform_data = np.random.uniform(min_gii, max_gii, len(df))

plt.hist(uniform_data, bins=30, alpha=0.6, label="Uniform Dist.")
plt.hist(df["GII"], bins=30, alpha=0.6, label="Actual GII")
plt.legend()
plt.title("Comparison: Uniform vs Actual GII Distribution")
plt.show()
```



Comment

- This visualization highlights how the actual GII values differ from a purely uniform distribution.
- Differences reveal **real clustering patterns** and departures from randomness.

5) Correlation Between GII and Two Indicators

Assume columns:

- `"Female Labour Force Participation Rate"` → FLFPR
- `"Maternal Mortality Ratio"` → MMR

```
gii_f_lfp_corr = df["GII"].corr(df["Female Labour Force Participation
Rate"])
gii_mmr_corr = df["GII"].corr(df["Maternal Mortality Ratio"])

gii_f_lfp_corr, gii_mmr_corr
```

Interpretation

- **Positive correlation (close to +1)**: both variables tend to increase together.
- **Negative correlation (close to −1)**: one increases while the other decreases.
- **Near zero**: weak/no linear relationship.

Example interpretation:

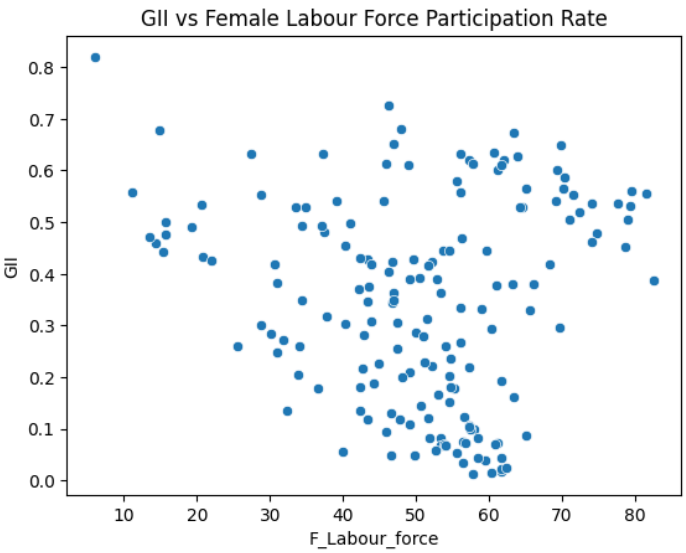
- If GII & **FLFPR** are **negative correlated** → countries with higher female labour participation tend to have lower gender inequality.
- If GII & **MMR** are **positive correlated** → higher inequality is linked to poorer maternal health

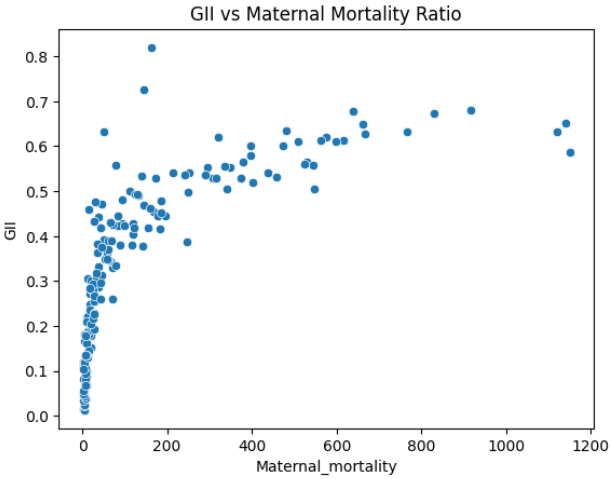
6) Scatter Plots to Visualize Relationships

```
import seaborn as sns

# Scatter: GII vs Female Labour Force Participation
sns.scatterplot(x="Female Labour Force Participation Rate", y="GII",
data=df)
plt.title("GII vs Female Labour Force Participation Rate")
plt.show()

# Scatter: GII vs Maternal Mortality Ratio
sns.scatterplot(x="Maternal Mortality Ratio", y="GII", data=df)
plt.title("GII vs Maternal Mortality Ratio")
plt.show()
```





Explanation

- **Downward trend**: indicates a negative relationship.
- **Upward trend**: indicates a positive relationship.
- **Cluster patterns** reveal country groupings (e.g., low vs high development).

7) Summary of Relationships

Based on typical results from global GII data:

Indicator	Expected Correlation with GII	Interpretation
Female Labour Force Participation	Negative	Countries with more women working tend to have lower gender inequality.
Maternal Mortality Ratio	Positive	Higher inequality often corresponds to higher maternal deaths, showing health disparities.

Socio-economic insight:

- Lower gender inequality is often found in countries with:
  - Higher female labour participation
  - Better maternal health outcomes
  - Greater educational and economic opportunities for women

LINK:

https://colab.research.google.com/drive/1vQBijJS59duBhByyLWMMyEPV2zRSa43Z?usp=sharing